DSA CASE STUDY

ViewSync Platform

Streaming Video Network Syndication Mesh

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Data Structures & Algorithms with C++

Problem Statement

Slow Catalog Classification

Organizing millions of video titles is inefficient and unscalable.

No Viewing State Rollback

Subscriber state changes cannot be tracked or reversed on error.

Random Task Execution

Profile processing tasks pile up with no guaranteed order.

Slow License Lookup

Finding a publisher trademark license takes too long.

No Delivery Optimization

There is no map of global licensing platforms to find lowest-cost delivery paths.

Objectives

1. Apply Data Structures and Algorithms to a real-world application
2. Implement efficient content catalog management
3. Manage viewing history with rollback functionality
4. Process system tasks using queue scheduling
5. Verify content licenses using hash-based lookup
6. Rank clips based on viewer retention
7. Model content syndication networks using graphs
8. Compute shortest delivery routes using Dijkstra's Algorithm
9. Balance encoding workloads using priority queues
10. Develop a menu-driven C++ application using STL

System Architecture

1 Catalog Directory	2 Correction Log	3 Algorithm Pipeline	4 Registration Code
5 Clip Sorter	6 Syndication Grid	7 Delivery Transport	8 Frame Splitter

Data Structures & Algorithms Used

Catalog Directory

Vector, Unordered Map

Correction Log

Stack

Algorithm Pipeline

Queue

Registration Code

Unordered Map

Clip Sorter

STL Sort — O(n log n)

Syndication Grid

Graph (Adjacency List)

Delivery Transport

Dijkstra's Algorithm

Frame Splitter

Priority Queue (Min Heap)

Module Implementation

Catalog Directory

Vectors + Unordered Maps for O(1) average search by ID, genre, or tag.

Correction Log

Stack stores viewing state changes; pop() gives instant undo on error.

Algorithm Pipeline

Queue ensures FIFO processing of profile updates and recommendations.

Registration Code

Hash map keyed by license code allows constant-time region verification.

Clip Sorter

STL sort on retention values surfaces highest-performing clips first.

Syndication Grid

Adjacency list graph models platform relationships and licensing links.

Delivery Transport

Dijkstra's Algorithm computes minimum-cost route between content providers and end users.

Frame Splitter

Priority queue assigns workloads to the least-loaded processor for balanced scheduling.

Complexity Analysis

Operation	Time Complexity	Space Complexity
Add Video	O(1)	O(n)
Search Video	O(1) Average	O(n)
Record Viewing State	O(1)	O(n)
Undo Viewing State	O(1)	O(n)
Add Pipeline Task	O(1)	O(n)
Process Pipeline Task	O(1)	O(n)
Verify License	O(1) Average	O(n)
Sort Clips	O(n log n)	O(n)
Add Graph Edge	O(1)	O(V + E)
Dijkstra's Algorithm	O((V + E) log V)	O(V)
Frame Scheduling	O(n log p)	O(n)

Legend: n = no. of elements | V = no. of vertices (platforms) | E = no. of edges (links) | p = no. of processors

Sample Execution Output

Video Catalog

ID	TITLE	GENRE	CATEGORY
101	Ocean Crimes	Thriller	Series
102	Galaxy Kids	Animation	Movie
103	Chef World	Reality	Show

License Verification

```
Code   : LIC-NET-IND
Region : India
Result : License Verified ✓
```

Clip Sorter

Rank	Clip Title	Ret.
1	Chef World Finale	9.50
2	Ocean Crimes Opening	8.70
3	Galaxy Kids Trailer	6.20

Delivery Route

```
Source : Rights Holder
Dest   : User Device
Cost   : 25
Route  : Holder→India CDN→Device
```

Frame Splitter

```
Processor 1 → Load: 14
Processor 2 → Load: 13
Processor 3 → Load: 10
Total Time  : 14
```

Results & Findings

- ✓ Hash-based indexing enables O(1) average content and license lookup
- ✓ Stack operations allow quick rollback of viewing state changes
- ✓ Queue scheduling guarantees FIFO task execution for consistency
- ✓ STL sorting effectively ranks clips by viewer retention
- ✓ Graph structures accurately model content distribution networks
- ✓ Dijkstra's Algorithm successfully computes optimal delivery routes
- ✓ Priority queues efficiently balance encoding workloads
- ✓ Multiple DSA concepts work together within a single integrated application

Conclusion

The ViewSync Platform successfully demonstrates the practical application of Data Structures and Algorithms in a streaming platform environment.

- → Vectors and Hash Maps for fast catalog management
- → Stack-based Correction Log for reliable undo operations
- → Queue-driven Pipeline for ordered task scheduling
- → Graph + Dijkstra for optimal content delivery routing
- → Priority Queue for balanced encoding workload distribution

GitHub: **github.com/sanikakangane/VIEWSYNC-PLATFORM**